

Baseline Correction

1) Objectives of Baseline Correction

A Notch Filter is a **band-stop filter** applied to EEG data. It:

- Targets a very narrow frequency band (typically 50 Hz or 60 Hz)
- Attenuates signals at that specific frequency
- Leaves frequencies just above and below mostly unchanged

2) How It Works

Baseline Correction:

- Selects a pre-stimulus interval (typically **-200 ms to 0 ms** before stimulus onset)
- Computes the mean amplitude of that interval for each trial
- Subtracts that mean from **the entire trial** (both pre- and post-stimulus data)

Result: Each trial is centered near zero during the baseline period.

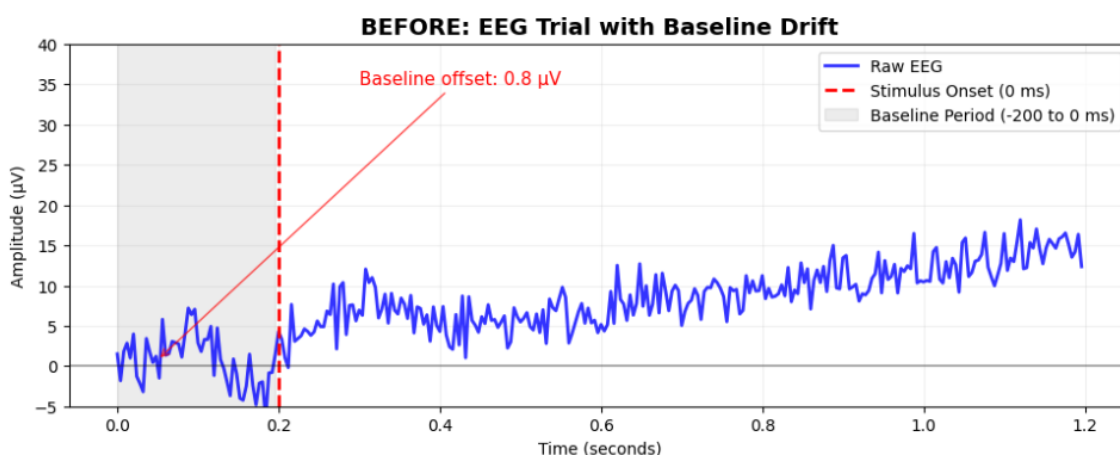


Figure 1 — Single EEG trial before baseline correction

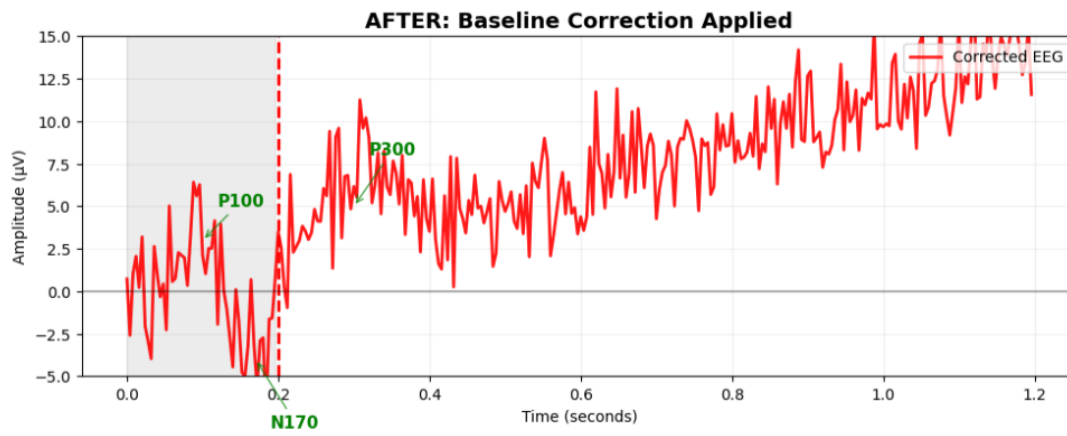


Figure 2 — Single EEG trial after baseline correction

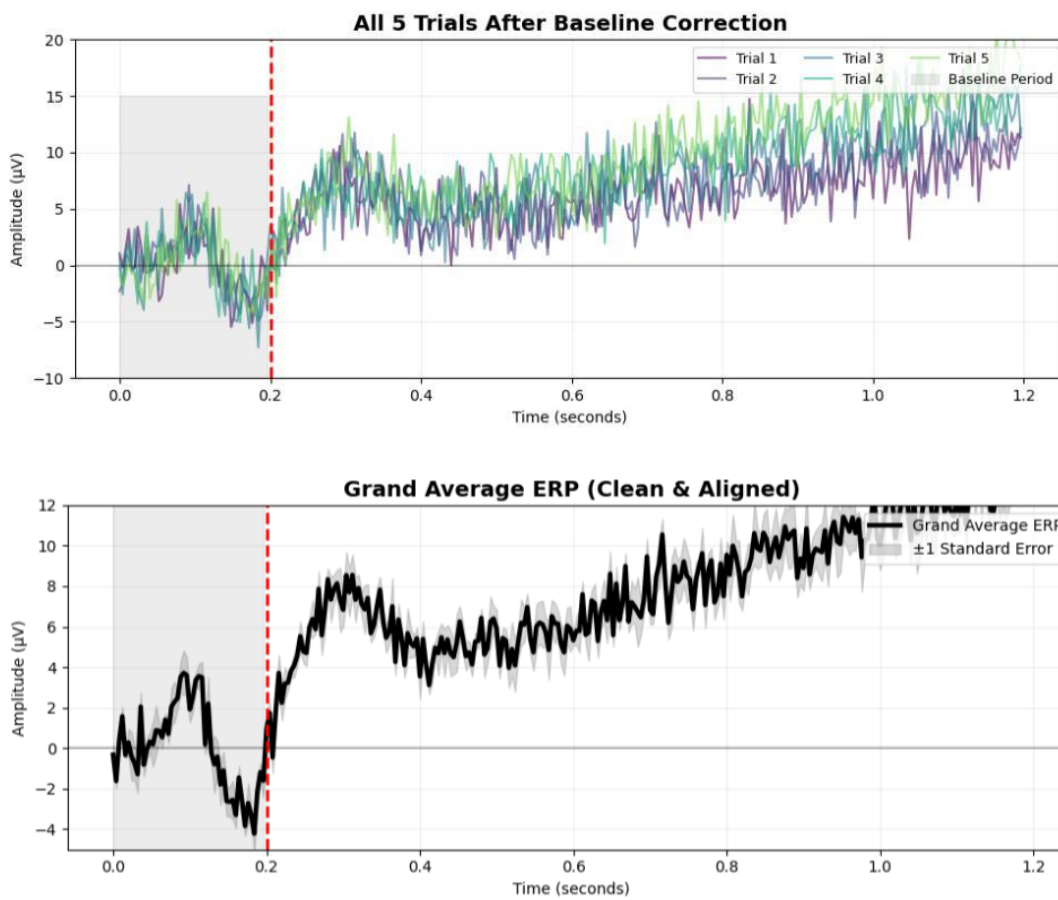


Figure 3 — All trials and grand average ERP after baseline correction